

Introduction to supercomputing

Houston Area Biophysical Simulation Workshop

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The age of computing



1964: CDC 6600

“First” supercomputer

Capable of 1 megaFLOP (1 million)

FLOPs= operations (+,-,X) per second



2014: iPhone 6

~1000 megaFLOPS

2022: iPhone 13

~768,000 megaFLOPS

How large are machines now?

• ■

1 Pf machine
NEU "Discovery"
~30000 cores

~several Tf GPU

iPhone

Area of each signifies
the performance

~200 Pflop machines (2015)

Titan

Sequoia

JUGEEN

MilkyWay-2

Sunway TiahuLight

The current leader (2022)



Frontier Supercomputer at ORNL

More than 600,000 CPU cores and 37,000 GPUs

Fiber optic connections between the cores

Capable of 1,000,000,000,000,000,000 calculations per second

~10,000,000,000,000 times more powerful than the CDC 6600!!!!

Weighs approximately 200,000 kg!

What does this mean?

First teraflop machine appeared in 1996 (ASCI “Red”)

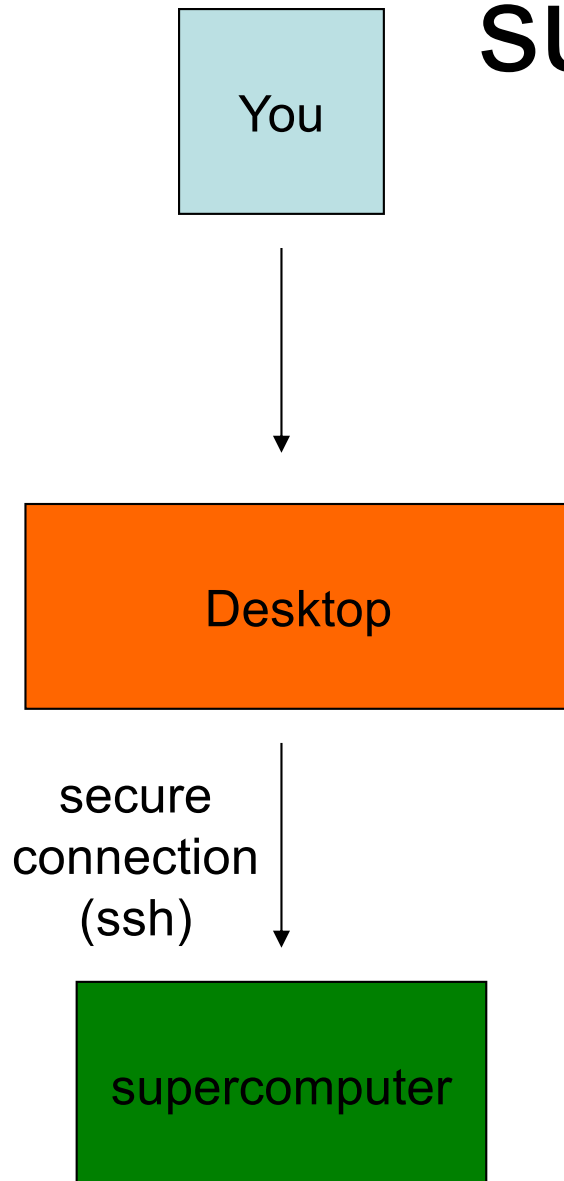
A single GPU can exceed 1 teraflop (~\$2000)

The supercomputing of the early 90s is trivial today

The cost of computation is small enough that anyone can afford to incorporate computational components into their research

**Computers are only tools, we need to ask
the right questions!!!**

How do you use a supercomputer?



```
pcw2@bgq-fn:~  
Problems and/or questions should be submitted via the problem tracking  
system form: http://helpdesk.rice.edu  
Enter remote host: bgq  
connecting to bgq  
pcw2@bgq's password:  
Last login: Mon Mar 16 14:16:13 2015 from   
-----  
RICE UNIVERSITY BLUEGENE/Q  
=====
```

Unauthorized access is prohibited.

BGQ,RICE,EDU

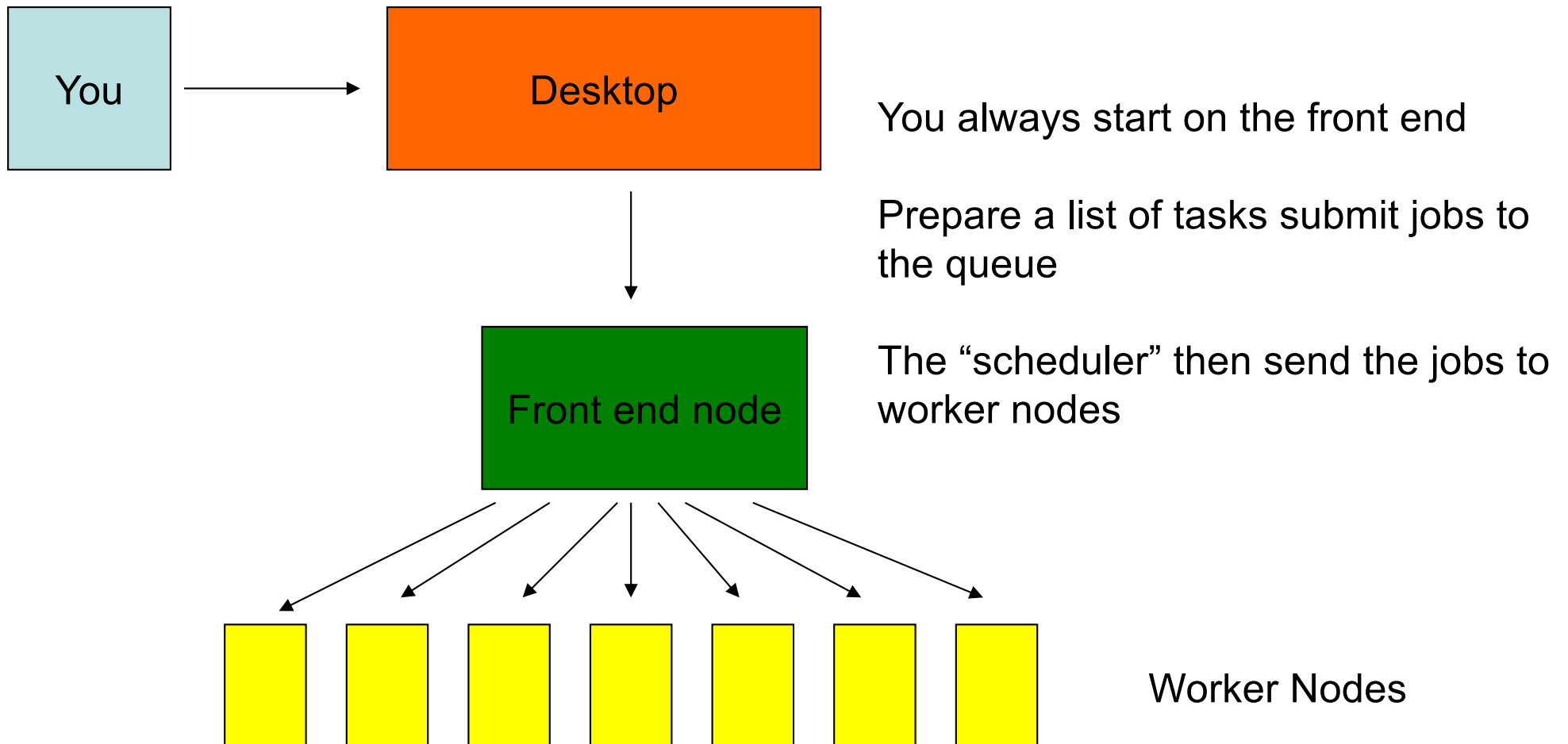
Questions and Problem Reports -> <http://helpdesk.rice.edu>

Welcome to Bluegene/Q, please use the "module" command to obtain a list
of supplied compilers, tools and specialized libraries.

Thanks,

RCSG.
[pcw2@bgq-fn ~]\$

Where is the work done?



The point?

- Using a command line is the only real option for using high performance computing resources
- We will use a terminal to connect to a compute cluster and will perform the simulations on GPUs within that machine

Using a Terminal

The terminal is based on **commands**.

You can't click on things!

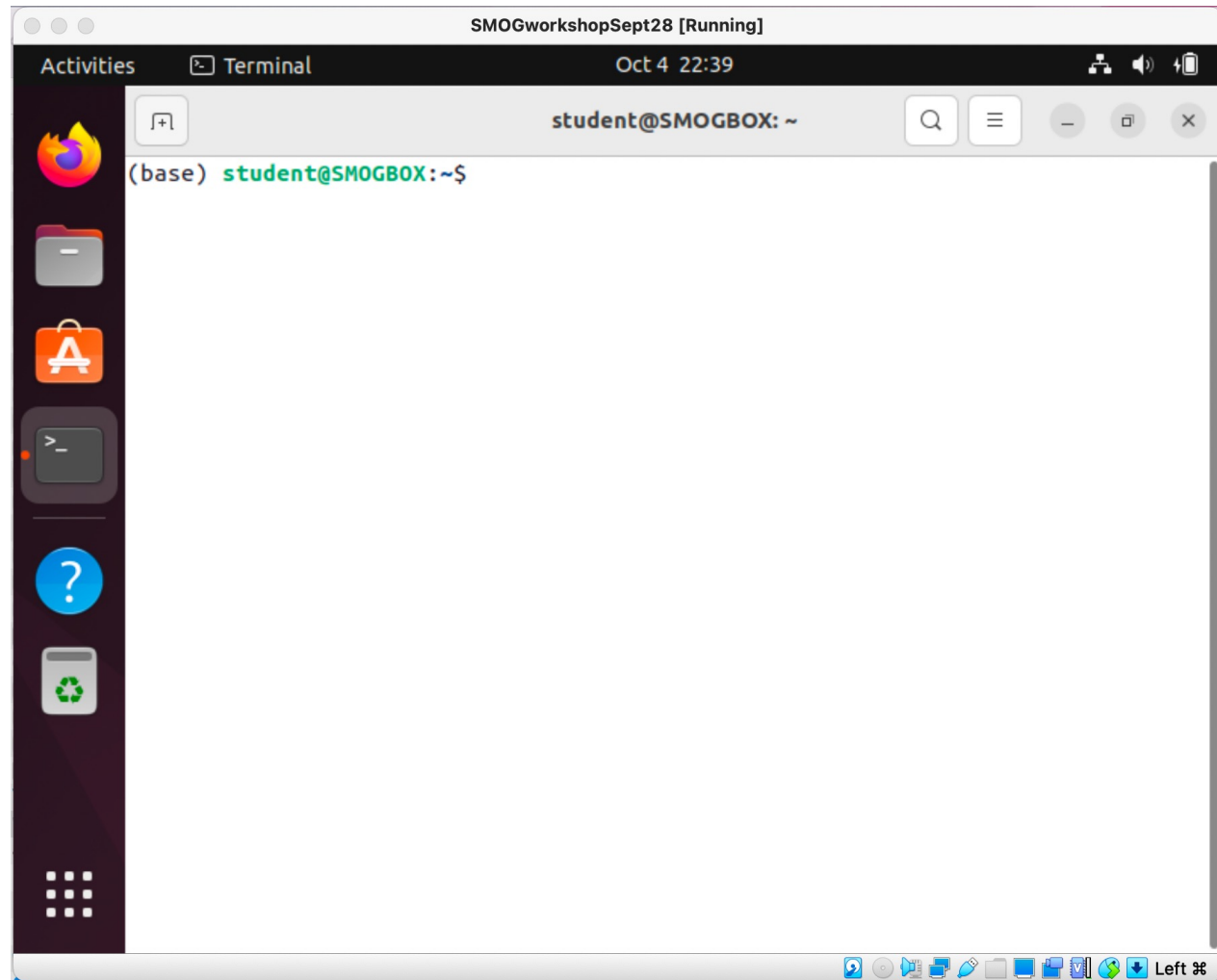
In these notes, commands will be shown in
COURIER font

Opening the terminal

- On your Windows machine, use the search bar at the bottom and type “powershell”
- Select “Powershell”
- This will open a terminal

You are now in the terminal

Your terminal will look a bit different, but the main point is that there is a command line.



Trying useful Linux commands

- Linux is command based
 - i.e. You enter text (commands) and hit “enter” key if you want something to happen
- Common Commands (follow along)
 - Change directory (going into ~/Desktop)
 - command : `cd Desktop`
 - List contents of directory
 - Command : `ls`
 - Get out of directory
 - command : `cd ..`
 - Show where you are :
 - Command : `pwd`

Connect to the supercomputer

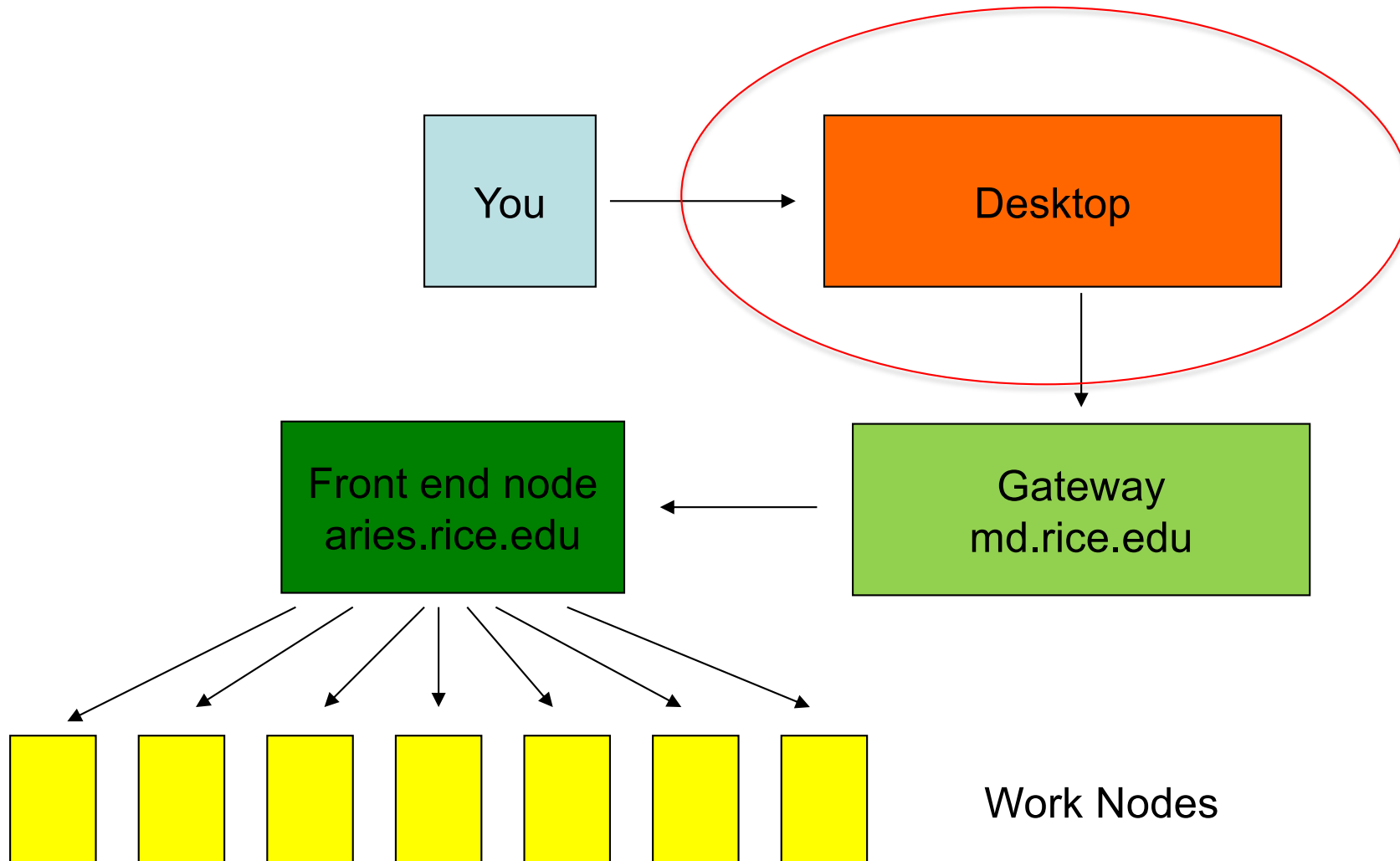
- First, establish a connection with a gateway node (`md.rice.edu`)
- Second, connect to the supercomputer
- Third, connect to a compute node of the cluster
- Fourth, run our calculations

Connect to the gateway

command: `ssh <username>@md.rice.edu`

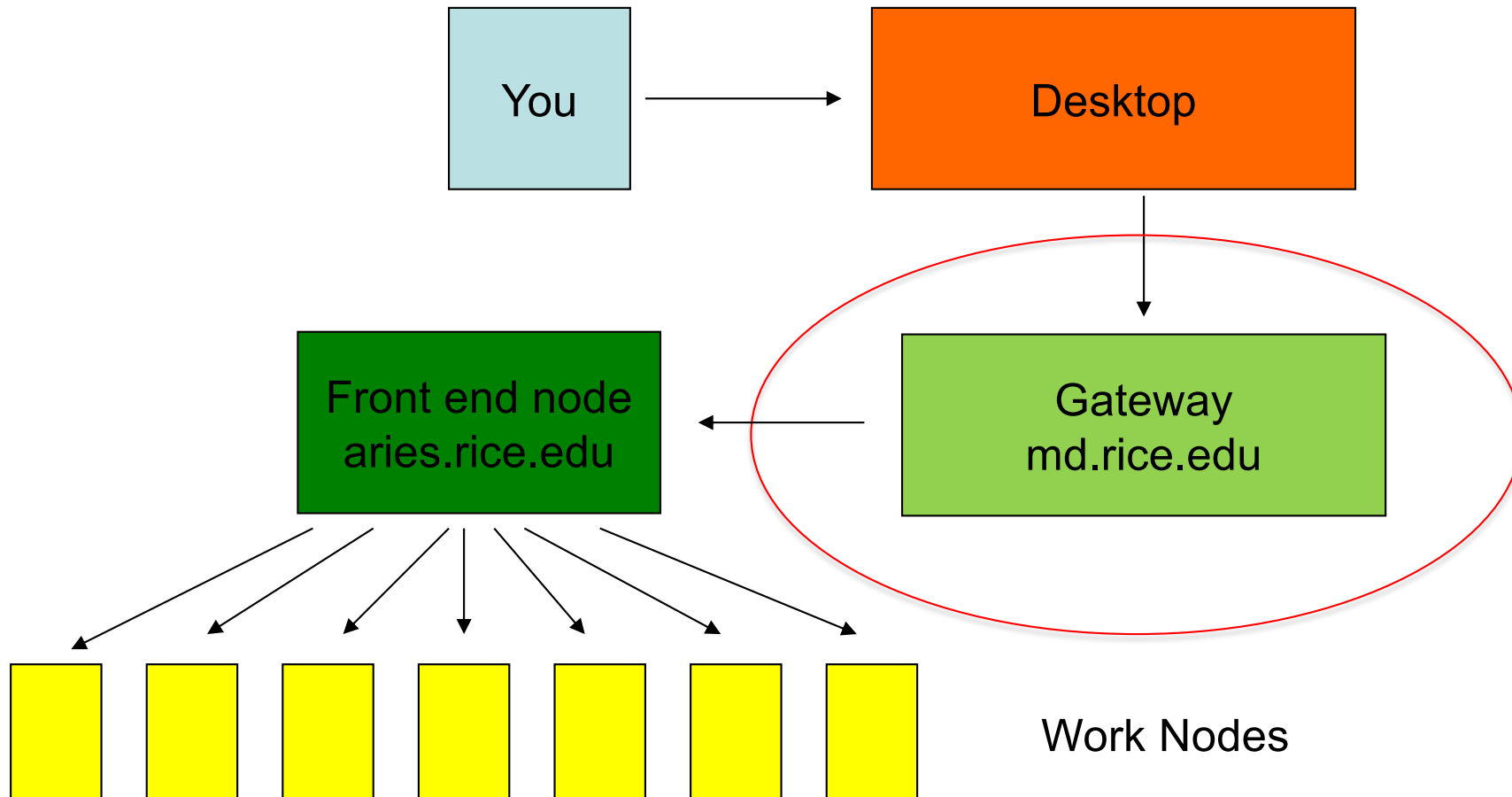
Your username is “hpc”, followed by an integer.

For example : `ssh hpc8@md.rice.edu`



Connect to the cluster

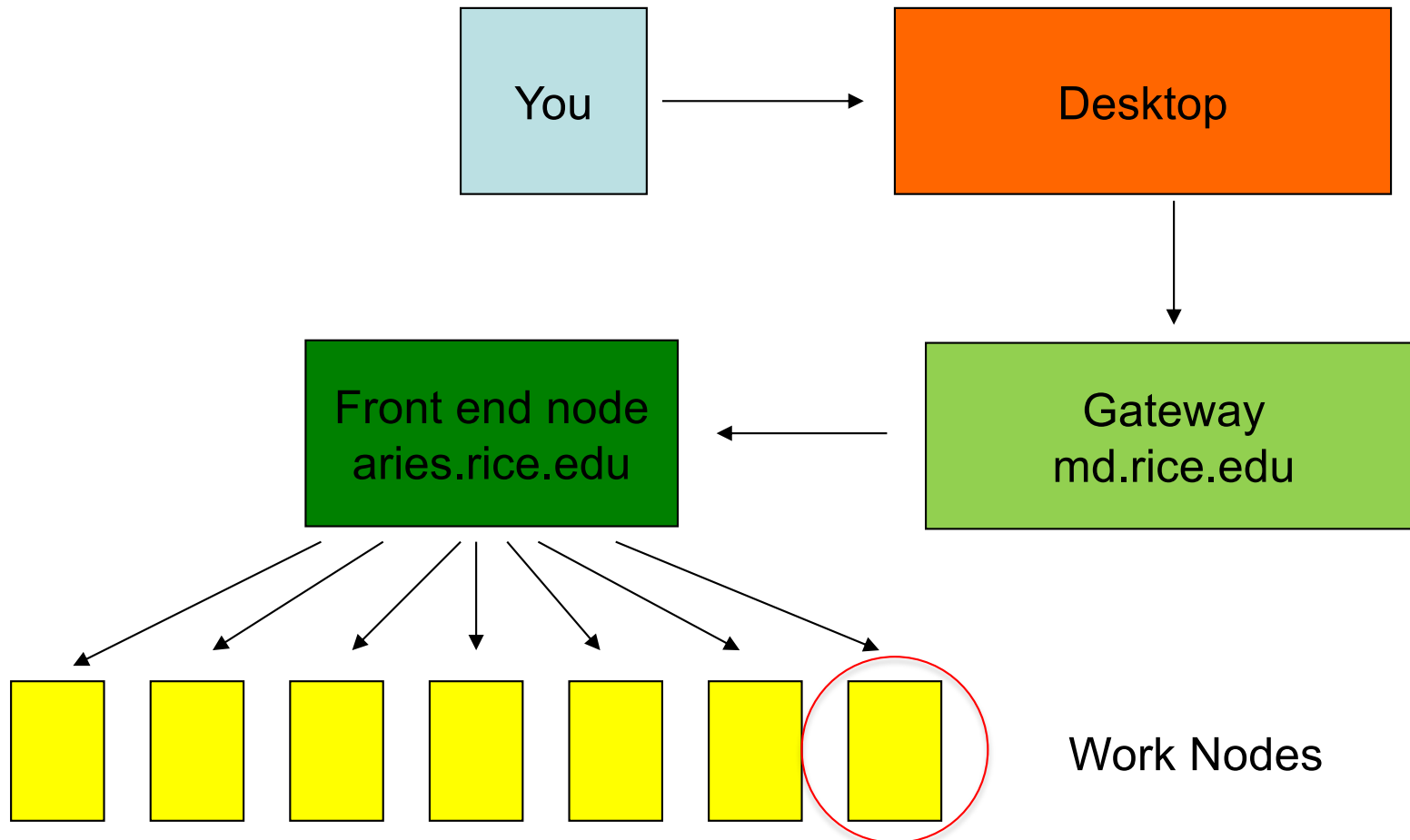
command: `ssh aries.rice.edu`



Get on a compute node

command: `bin/slurmJob.sh`

This will reserve one GPU for you!



Launch a “container”

A “container” is a package of software that contains all of the elements required to run a specific type of calculation

We prepared a container for this workshop.
Launch it with the command :

```
singularity shell /home/hpc1/ctbp.sif
```

Let's go!

- We are now on a compute node with our own GPU and all programs/libraries accessible. It's time to rock and roll!